



IT3100 – AI APPLICATION PROJECT

Project Guidebook

2026 S1

School of
Information
Technology

1. Introduction to Learning Unit (LU)

This project LU allows students to work on an Artificial Intelligence (AI) project. You will make use of design thinking and its application in the problem-solving process, the Cross-Industry Process for Data Mining (CRISP-DM) for collecting and processing data, selecting and evaluating AI models, developing Gen AI capabilities and finally deploying and integrating them with the application you have developed.

Working in a team, you will apply the knowledge and skills that you have learned in the other AI LUs in this specialization to solve a real-world problem.

2. Learning Outcomes

On the successful completion of this LU, you will be able to:

1. Describe the challenges and constraints faced in applying AI solutions.
2. Describe the use of AI-enabled cloud services used in the enterprise.
3. Apply the appropriate machine learning tools and platforms to build AI applications that solve real world problems.
4. Plan and execute a project as a team iteratively and deliver the solutions in a timely manner according to the given requirements.

3. Project Organisation

Project teams will consist of 3 to 4 students. Each team must elect a team leader.

Role of Team Leaders

- Coordinate and assign work among members.
- Ensure that the team is complying with project requirements stated in the project guidebook.
- Keep track of the project schedule and ensure that the scheduled tasks are completed on time.
- Arrange meetings with project supervisor; this includes the booking of meeting venues
- Communicate with the supervisor and external party on behalf of the team as and when necessary

Role of Team Members

- Participate actively in all activities during practical and tutorial

- Cooperate with the rest of the team and complete all the tasks allocated. (Marks may be deducted for individuals who rely heavily on others to complete his/her tasks. This is to cultivate an independent working attitude.)
- Be present and punctual for all group meetings
- Respect the authority and decisions of the team leader

Project Supervisors (LU Tutor)

- Provide feedback on project teams' proposals and designs
- Advise the project team during project development, on matters including task allocation, tools used, etc.
- Provide inputs from the company management/user perspective

4. Project Details

Develop an ethical AI application that solves a real problem to benefit the lives of Singaporeans, under one of the following themes:

- Business
- Education
- Healthcare
- Logistics
- Smart Nation
- Sustainability
- Transport

Teams can research into problem statements surfaced in recent or current SG challenges, hackathons or competitions and choose/adapt one to work on. Using Design Thinking processes, you will then seek to understand your target users, discover what they would like to see in your solution, brainstorm solutions in line with the theme before proposing a solution that fits the theme. As a team, you will subsequently apply what you have learned in this or other learning units to collect and process data, train AI supervised learning models, develop Gen AI capabilities and integrate them into the application to showcase your proposed solution.

5. Scope of Work

Each team member is required to

1. Collect and understand the data and ensure that the data is prepared in a way that is readily understood by the Machine Learning (ML) model.

2. Train a classical or deep supervised learning model, conduct hyperparameters tuning and use suitable metrics (such as accuracy, precision, recall) to evaluate the performance.
3. Document the data collection, understanding, preparation and model training experiments.
4. Develop a Gen AI capability with clearly defined use cases.
5. Develop an application that has your ML model and Gen AI capability integrated.
6. Develop additional features (e.g. user management, interface to API services like Maps API, mobile app, etc.) to enhance your team solution.
7. Present your work in detail.

6. Tasks & Schedule

Week	Tasks	Individual (%)	Group (%)
2	Administrative Tasks - Finalise teams - Draft problem statement	–	–
5	Idea Presentation - Project Plan - Slides	15	5
8	Progress Review 1 - Supervised learning model - Code Review - Team Progress	20	5
14	Progress Review 2 - Gen AI capability - Code Review	20	–
18	Final Presentation - Final Report - Slides - Source Codes - Peer Review - Ethical AI Checklist	30	5

7. Project Supervisor

LU Leader & Tutor	Lim Sing Tat
------------------------------	--------------

Contact	6550 1731 lim_sing_tat@nyp.edu.sg
----------------	--------------------------------------

8. Annex A – Suggested content

1. Idea Presentation

- a. Project Plan (Individual)
 - Scope of work
 - Supervised Learning Model
 - Gen AI Capability
 - Milestones & Deliverables
 - Reference links (for any research done)
- b. Slides (Group)
 - Problem statement
 - Research into problem
 - Individual component
 - Schedule

2. Final Report

- a. Solution
- b. Models
 - Supervised Learning Model
 - Gen AI Capability
- c. Features
- d. UX
- e. Reflection

Rubrics

1. Idea Presentation

Individual					
Criteria	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Good (B)	Excellent (A)
Presentation [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Verbal & Non-verbal Communications	• Voice is consistently too weak or too strong • Fails to use inflections to emphasise key points or to create interests • Appears very uneasy and insecure • Lacks or has body motions or gestures that are highly distracting	• Voice is frequently too weak or too strong • Rarely uses inflections to emphasise key points or to create interests • Appears uneasy and somewhat insecure • Uses few body motions or gestures that distract audience	• Voice is generally strong • Sometimes uses inflections to emphasise key points • Appears generally at ease and confident • Body motions and gestures are neutral	• Voice is strong • Often uses inflections to emphasise key points • Appears mostly comfortable and confident • Uses body motions and gestures well	• Voice is confident and strong • Consistently uses inflections to emphasise key points • Appears very comfortable and confident • Uses body motions and gestures very effectively
Q & A	• Unable to respond to questions or provide irrelevant answers	• Able to respond with somewhat relevant, vague answers	• Able to respond with relevant, direct answers	• Able to respond with relevant answers and short elaborations	• Able to respond with relevant answers and insightful elaborations
Scope of work					

[10 Marks]		Marks: 0 to < 5	Marks: 5 to < 6	Marks: 6 to < 7	Marks: 7 to < 8	Marks: 8 to 10
Proposed Supervised Model & Gen AI capabilities		None or does not address problem statement	Addresses problem statement but contains some irrelevant details	Addresses problem statement and contains relevant details with brief explanations	Addresses problem statement with clear indication of AI elements and some explanations	Addresses problem statement with clear indication of AI elements and excellent explanations and visuals
Content		None or poorly prepared with no details on scope of work to achieve key milestones and deliverables	Briefly prepared with many missing details on scope of work required to achieve key milestones or deliverables	Partially prepared with some missing details on scope of work required to achieve key milestones or deliverables	Prepared with minor missing details on scope of work required to achieve key milestones and deliverables	Well-prepared with clearly defined and team-aligned scope to achieve all key milestones and deliverables
Resource		Poorly planned with most workload handled by one member	Briefly planned with workload distributed across less than half the team	Moderately planned with workload distributed across half the team	Well planned with workload distributed across more than half the team	Skillfully planned with workload distributed evenly across all members
Group						
Presentation [5 Marks]		Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Slides		None or irrelevant	Content is relevant but fragmented and poorly organised	Content is relevant and moderately organised	Content is relevant, well organised and coherent	Content is relevant, well organised, coherent and extensive

Execution	• Complete lack of preparation resulting in incoherent and confusing execution	• Poor preparation between members resulting in major execution problems	• Moderate preparation between members resulting in minor execution problems	• Good preparation between members resulting in minimal execution problems	Excellent preparation between members resulting in excellent execution
-----------	--	--	--	--	--

2. Progress Review 1

Individual					
Criteria	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Good (B)	Excellent (A)
Input Data [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Collection	• No data or irrelevant data collected	• Data collected is relevant but only sufficient for use with pre-trained models	• Data collected is relevant and marginally sufficient from a single source	• Data collected is relevant and substantial from a single source	• Data collected is relevant, substantial and coherent; process is sophisticated
EDA/Preparation	• No EDA done	• Minimum EDA done and many issues with data preparation	• Moderate EDA done and some issues with data preparation	• Good EDA with simple explanations and no issue with data preparation	• Excellent EDA with insightful details, use of visualisation resulting in comprehensive data preparation
Supervised Model [10 Marks]	Marks: 0 to < 5	Marks: 5 to < 6	Marks: 6 to < 7	Marks: 7 to < 8	Marks: 8 to 10
Training	• None or incomplete	• Completed using only pre-trained	• Completed with custom training	• Completed with custom training	• Completed with custom training; no

Evaluation	• None or inappropriate use and interpretation of performance metrics	models without subsequent training • Appropriate use and interpretation of performance metrics with major misconceptions	and major conceptual errors • Appropriate use and interpretation of performance metrics with moderate misconceptions	and minor conceptual errors • Appropriate use and interpretation of performance metrics with minor misconceptions	conceptual errors and evidence of optimisations • Appropriate use and in-depth interpretation of performance metrics with no misconceptions
Source Codes [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Structure	• None and no comments or markdown for explanations	• Poorly organised with confusing flows and poor use of comments or markdown	• Organised with logical flows with sparse or poor use of comments or markdown	• Organised with logical flows with and good use of comments or markdown	• Well-organised, logical, modular with excellent use of comments or markdown
Code Review	• Unable to explain critical components of the codes or provide clarifications when probed	• Able to explain limited critical component of the codes conveying sparse subject knowledge	• Able to explain some critical components of the codes conveying decent subject knowledge	• Able to explain most critical components of the codes conveying good subject knowledge	• Able to explain all critical components of the codes conveying excellent subject knowledge
Group					
Plan [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Progress	• No details on refining plan to	• Poor progress update with	• Some progress update with minor	• Good progress update with	• Comprehensive progress update

	address new requirements or changes in scope	significant gaps in revised plan to address changes in scope	gaps in revised plan to address changes in scope	minimal gaps in revised plan to address changes in scope	with no gaps and sound recommendations in revised plan to address changes in scope
--	--	--	--	--	--

3. Progress Review 2

Individual					
Criteria	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Good (B)	Excellent (A)
Supervised Model [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Optimisation	None or poor techniques resulting in accuracy < 50%	Basic techniques resulting in accuracy $\geq$ 50%	Intermediate techniques resulting in accuracy $\geq$ 60%	Good techniques resulting in accuracy $\geq$ 70% (or industry benchmark)	Systematic and iterative techniques resulting in accuracy $\geq$ 80% (or industry benchmark)
Deployment/Hosting	No model deployed	Model is deployed locally with major issues	Model is deployed locally with minor issues	Model is deployed locally without issues	Model is deployed remotely without issues
Gen AI Capability [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Use case and scope of work	Unclear objectives with poorly defined	Clear objectives with basic scope of	Clear objectives with intermediate	Clear objectives with good scope of	Clear objectives with excellent

Implementation	scope or no avenues to achieve them • None	work and research • Achieved with major errors or implementation gaps	scope of work and research • Achieved with moderate errors or implementation gaps	work and research • Achieved with minor errors or implementation gaps	scope of work and extensive research • Fully achieved with no errors or gaps, complete with detailed explanations
Application [5 Marks] Functionality	Marks: 0.0 to < 2.5 • Models not integrated leading to no results • No User Interface (UI) • No output	Marks: 2.5 to < 3.0 • Can showcase either supervised model or Gen AI capability • UI is totally not intuitive • Outputs do not illustrate problems are solved	Marks: 3.0 to < 3.5 • Can showcase both supervised model and Gen AI capability with moderate issues • UI is moderately intuitive • Outputs somewhat illustrate problems are solved	Marks: 3.5 to < 4.0 • Can showcase both supervised model and Gen AI capability with minor issues • UI is largely intuitive • Outputs largely illustrate problems are solved	Marks: 4.0 to 5.0 • Can showcase both supervised model and Gen AI capability with no issues • UI is very intuitive • Outputs clearly illustrate that problems are solved
Source Codes [5 Marks] Structure	Marks: 0.0 to < 2.5 • None and no comments or markdown for explanations	Marks: 2.5 to < 3.0 • Poorly organised with confusing flows and sparse use of comments or markdown	Marks: 3.0 to < 3.5 • Moderately organised with logical flows and moderate use of comments or markdown	Marks: 3.5 to < 4.0 • Largely organised with logical flows good use of comments or markdown	Marks: 4.0 to 5.0 • Well-organised, logical, modular with excellent use of comments or markdown

Code Review	• Unable to explain or clarify critical components of the codes when probed	• Able to explain minimum critical components of the codes conveying sparse subject knowledge	• Able to explain some critical components of the codes conveying moderate subject knowledge	• Able to explain most critical components of the codes conveying good subject knowledge	• Able to explain all critical components of the codes conveying excellent subject knowledge
-------------	---	---	--	--	--

4. Final Project Presentation

Individual					
Criteria	Inadequate (F)	Acceptable (D)	Satisfactory (C)	Good (B)	Excellent (A)
Presentation [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Delivery	• Voice is consistently too weak or too strong • Fails to use inflections to emphasise key points or to create interests • Appears very uneasy and insecure • Lacks or has body motions or gestures that are highly distracting	• Voice is frequently too weak or too strong • Rarely uses inflections to emphasise key points or to create interests • Appears uneasy and somewhat insecure • Uses few body motions or gestures that distract audience	• Voice is generally strong • Sometimes uses inflections to emphasise key points • Appears generally at ease and confident • Body motions and gestures are neutral	• Voice is strong • Often uses inflections to emphasise key points • Appears mostly comfortable and confident • Uses body motions and gestures well	• Voice is confident and strong • Consistently uses inflections to emphasise key points • Appears very comfortable and confident • Uses body motions and gestures very effectively
Q & A	• Unable to respond to questions or	• Able to respond with somewhat	• Able to respond with relevant,	• Able to respond with relevant	• Able to respond with relevant

	provide irrelevant answers	relevant, direct answers	direct answers	answers and short elaborations	answers and insightful elaborations
App Demo [10 Marks] Model, Gen AI Capability & Features User Experience (UX)	Marks: 0 to < 5 - Models not integrated into solution or unable to showcase features due to errors or failures - Poor usability without regards to any UX principles or human-centric design	Marks: 5 to < 6 - Models integrated into solution and able to showcase minimum features - Limited UX principles, low human-centric design and usability	Marks: 6 to < 7 - Models integrated into solution and able to showcase features with following attribute: (i) relevance - Basic UX principles, human-centric design and usability	Marks: 7 to < 8 - Models integrated into solution and able to showcase features with following attributes: (i) relevance (ii) tech proficiency - Moderate UX principles, human-centric design and usability	Marks: 8 to 10 - Models integrated into solution and able to showcase features with following attributes (i) relevance (ii) tech proficiency (iii) impact - Excellent UX principles, human-centric design and usability across solution
Ethical AI [5 Marks] Implementation Guidelines	Marks: 0.0 to < 2.5 Submitted relevant answers for < 50%	Marks: 2.5 to < 3.0 Submitted relevant answers for $\geq 50\%$	Marks: 3.0 to < 3.5 Submitted relevant answers $\geq 60\%$	Marks: 3.5 to < 4.0 Submitted relevant answers $\geq 75\%$	Marks: 4.0 to 5.0 Submitted relevant answers $\geq 90\%$
Reflection [5 Marks] Lessons Learned	Marks: 0.0 to < 2.5 No analysis with reference to project	Marks: 2.5 to < 3.0 Poor analysis with reference to project	Marks: 3.0 to < 3.5 Brief analysis involving narrow aspects of project	Marks: 3.5 to < 4.0 Good analysis involving narrow	Marks: 4.0 to 5.0 In-depth analysis involving broad

Improvements	No suggestions to improve the solution	Suggestions to improve solution are trivial	Suggestions to improve solution are relevant but lack details	aspects of the project Suggestions to improve solution are relevant with some details	aspects of the project Suggestions to improve solution are relevant, well-crafted and feasible
Peer Review [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Submission and reviews	Received extremely poor reviews from all members	Members highlighted major issues	Members highlighted moderate issues	Members highlighted minor issues	No issues highlighted by members
Group					
Presentation [5 Marks]	Marks: 0.0 to < 2.5	Marks: 2.5 to < 3.0	Marks: 3.0 to < 3.5	Marks: 3.5 to < 4.0	Marks: 4.0 to 5.0
Slides	None or irrelevant	Content is relevant but fragmented and poorly organised	Content is relevant and well organised	Content is relevant, well organised and coherent	Content is relevant, well organised, coherent and extensive
Execution	Complete lack of preparation resulting in incoherent and confusing execution	Poor preparation between members resulting in major execution problems	Moderate preparation between members resulting in minor execution problems	Good preparation between members resulting in minimal execution problems	Excellent preparation between members resulting in excellent execution